PROJECT PORTFOLIO

# Proposed 5-Storied Residential Building

Architectural and structural drawing set

22 drawings | Architectural, structural, utilities and location

| | |
|---|---|
| **OWNER** | Md. Humayun Kabir, S/O Md. Moynul Islam |
| **SITE** | Mouza Baraishal Para, Saidpur, Nilphamari, Bangladesh |
| **PLOT REFERENCE** | Khatiyan No. 16, Dag No. 6, J.L. No. 40 |
| **TOTAL LAND AREA** | 3.95 decimal |
| **GROUND FLOOR AREA** | 1,006 sft |
| **TOTAL FLOOR AREA** | 5,294 sft (as stated on the drawing sheet) |
| **CONSULTANT** | Landmark Engineering Solutions, Khoda Hafej Gate, Parbatipur Road, Saidpur, Nilphamari |
| **PREPARED BY** | Engr. Mobarok Hossain, B.Sc. Engg (WUB) |
| **CHECKED BY** | Engr. Saeed Sourav, B.Sc. Engg (Civil) |

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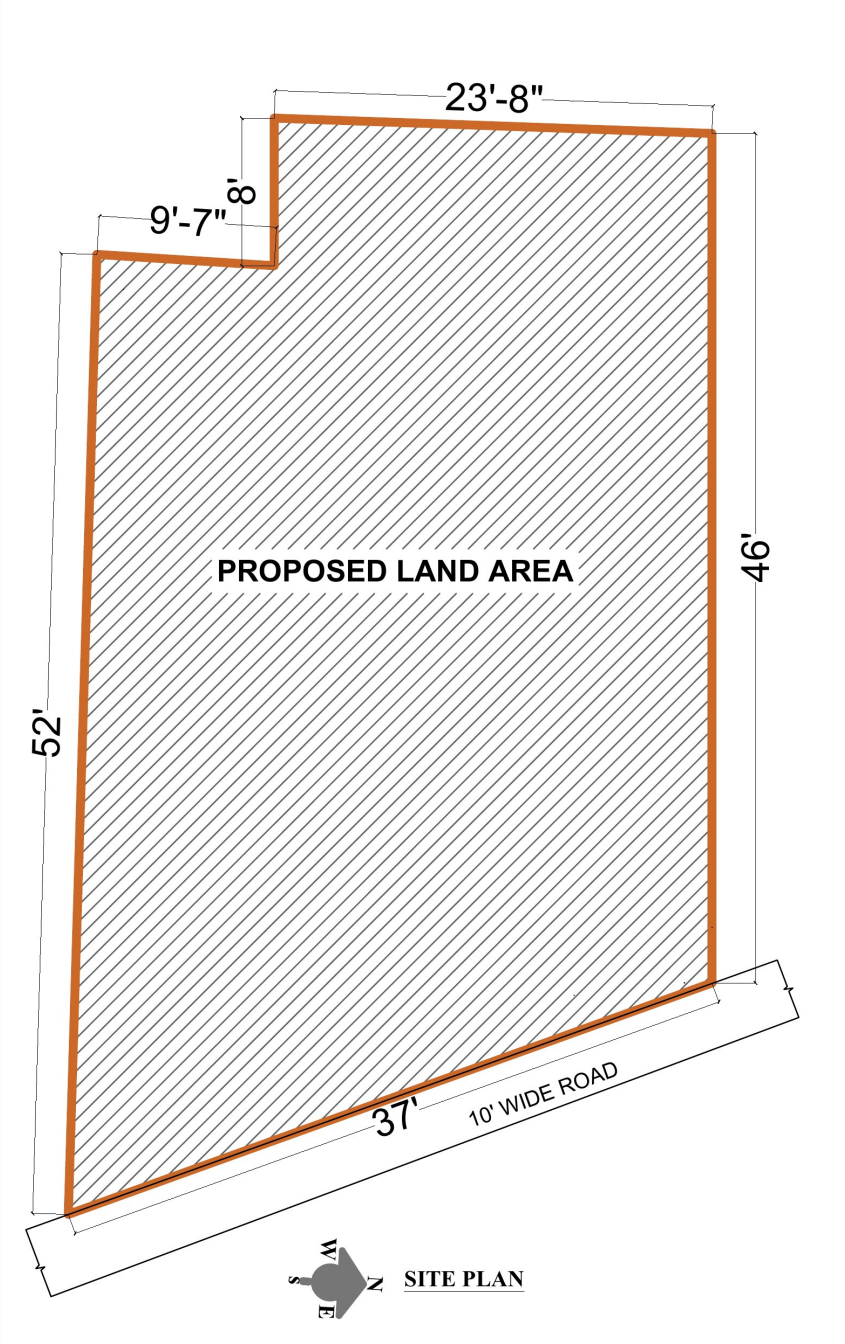
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# Site Plan



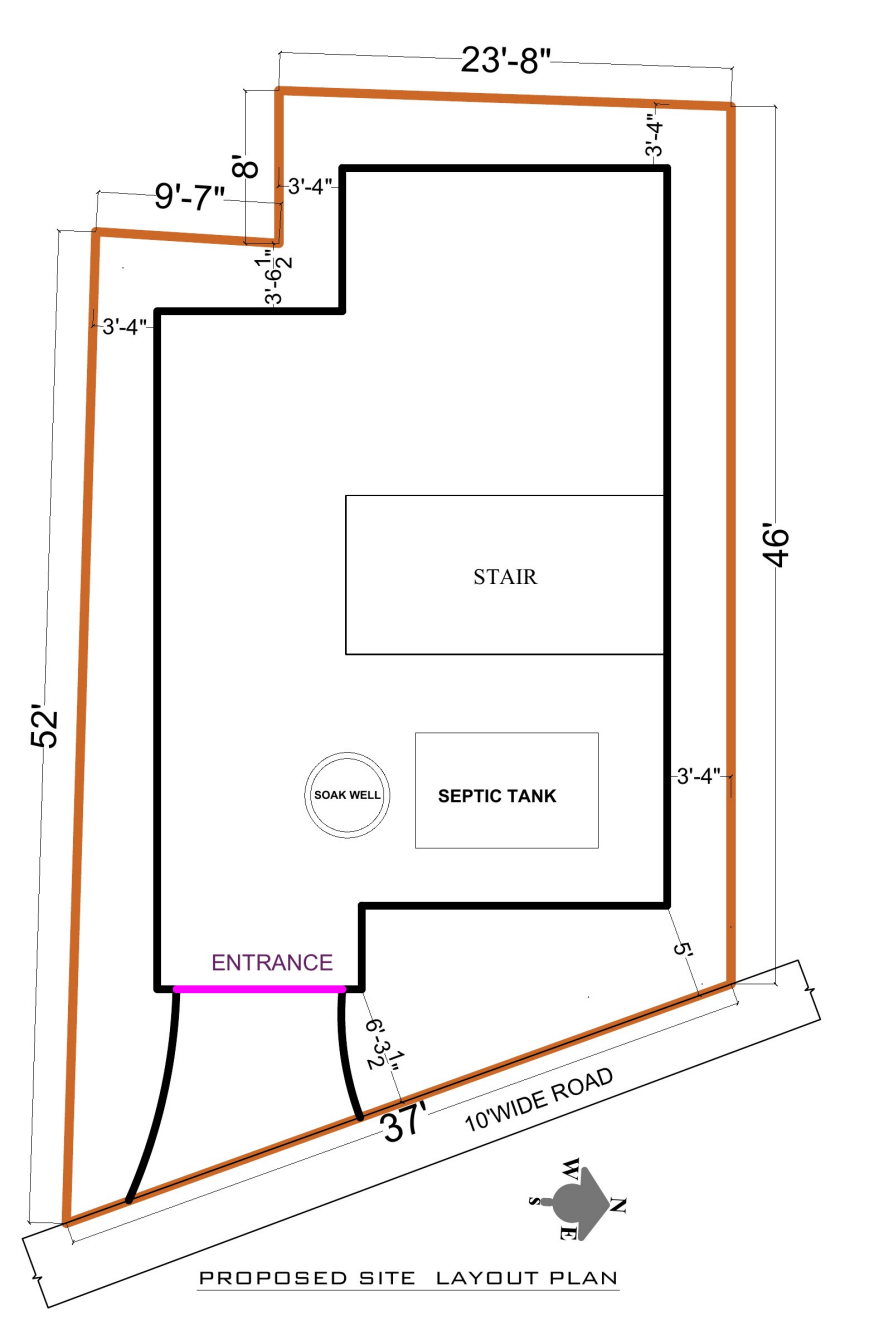

## DESCRIPTION

Shows the proposed land area (hatched), a small irregular plot on a 10' wide road. The plot outline is the basis for the building layout and setbacks.

## KEY INFORMATION FROM THE DRAWING

- Plot dimensions 23'-8" at the top, 46' and 52' on the sides, and 37' along the road
- Corner cut dimensions 9'-7" and 8' at the top-left
- Total land area 3.95 decimal (Khatiyan 16, Dag 6, J.L. No. 40)

# Proposed Site Layout Plan



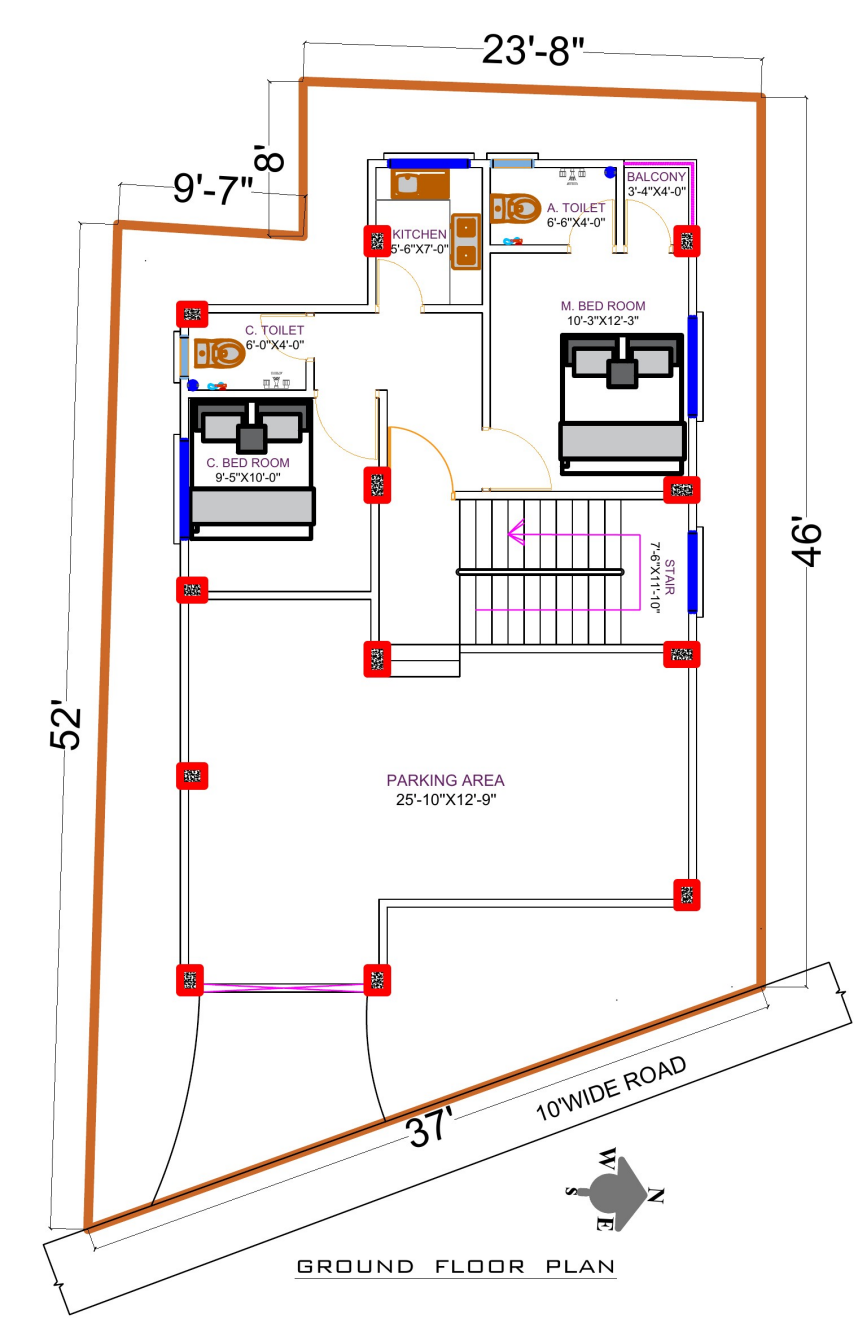

## DESCRIPTION

Places the building footprint on the plot and shows the entrance from the road, the stair, and the septic tank and soak well at the rear.

## KEY INFORMATION FROM THE DRAWING

- Entrance from the 10' wide road
- Setback dimensions of 3'-4" and 3'-6½" around the footprint
- Septic tank and soak well located within the plot
- Stair block of 7'-6" x 11'-10"

# Ground Floor Plan



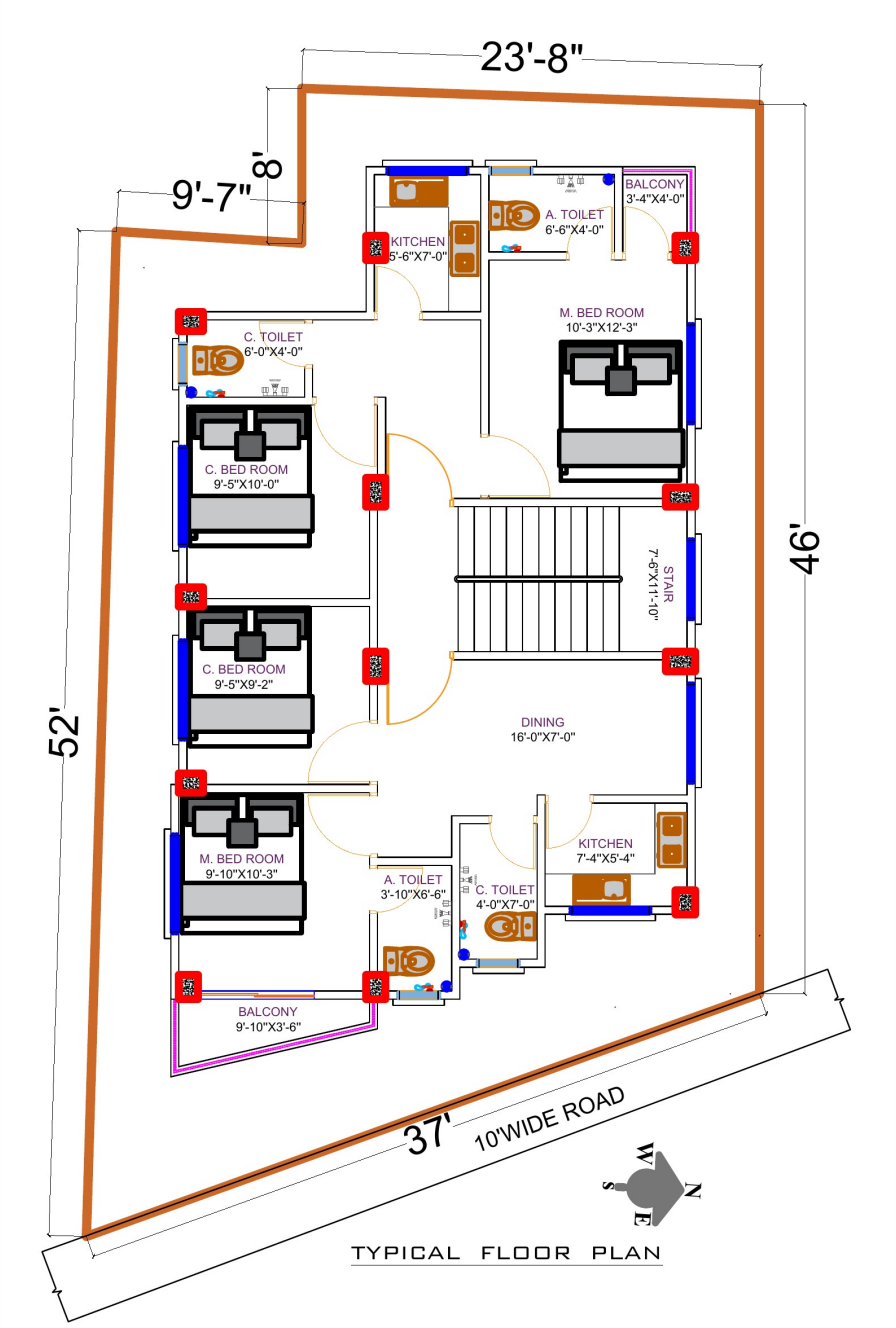

## DESCRIPTION

Furnished ground floor plan showing the parking area, the stair and the residential rooms, with their sizes marked on the plan.

## KEY INFORMATION FROM THE DRAWING

- Parking area 25'-10" x 12'-9"
- Stair 7'-6" x 11'-10"
- Master bedroom 10'-3" x 12'-3" and child bedroom 9'-5" x 10'-0"
- Kitchen 5'-6" x 7'-0", attached and common toilets, balcony 3'-4" x 4'-0"

# Typical Floor Plan



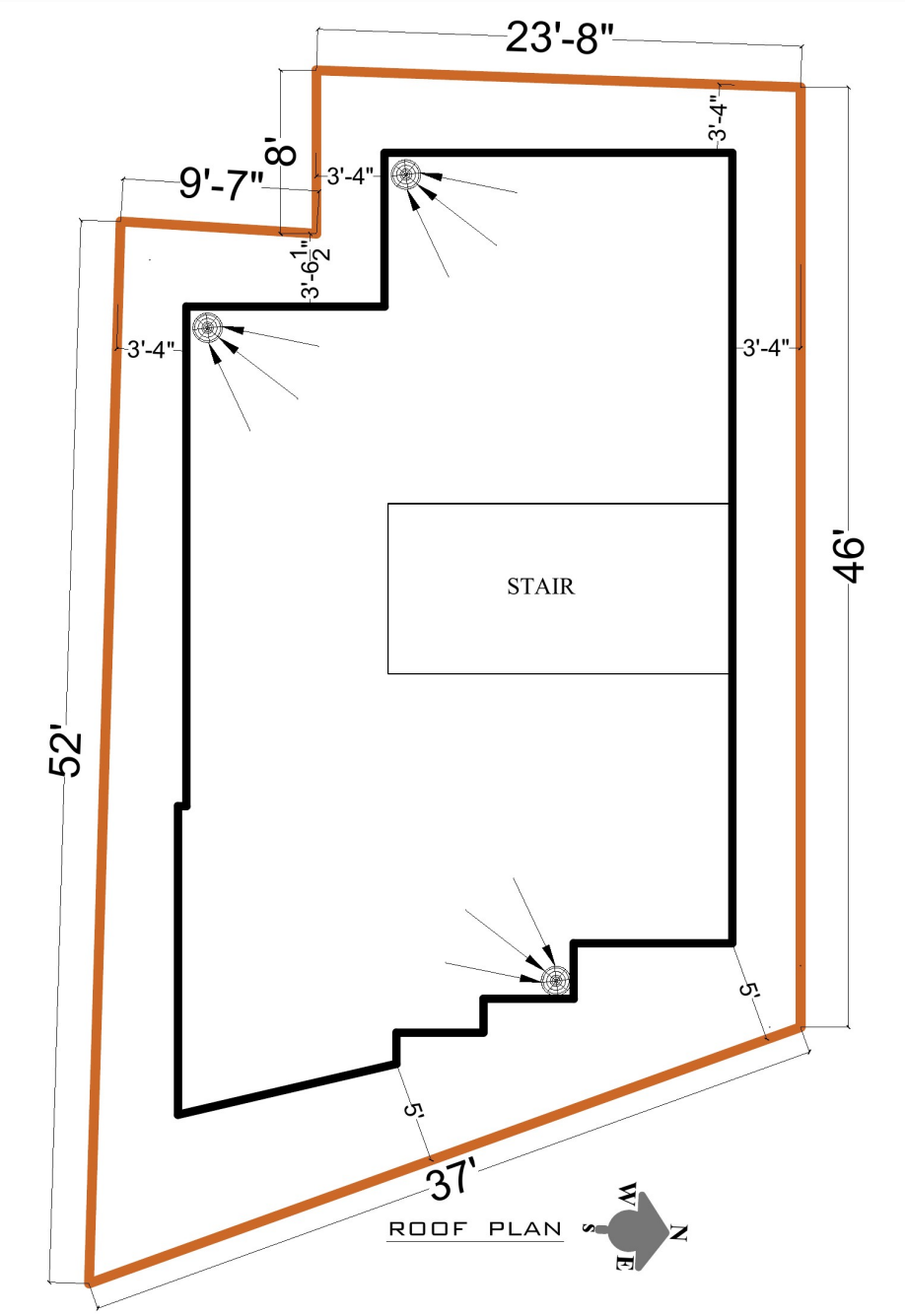

## DESCRIPTION

Furnished plan of the repeating floor used for the 1st to 4th floors. It holds two apartments, each with bedrooms, toilets, kitchen and balcony, served by the common stair.

## KEY INFORMATION FROM THE DRAWING

- Master bedrooms 10'-3" x 12'-3" and 9'-10" x 10'-3"
- Child bedrooms 9'-5" x 10'-0" and 9'-5" x 9'-2"
- Kitchens 5'-6" x 7'-0" and 7'-4" x 5'-4"
- Dining 16'-0" x 7'-0"; balconies 3'-4" x 4'-0" and 9'-10" x 3'-6"

# Roof Plan




## DESCRIPTION

Plan view of the roof slab with the stair head-room outline, drawn on the same plot outline as the layout plan.

## KEY INFORMATION FROM THE DRAWING

- Same plot outline and setback dimensions as the layout plan
- Stair head-room position marked
- Facing the 10' wide road

# Front Side Elevation



FRONT SIDE ELEVATION

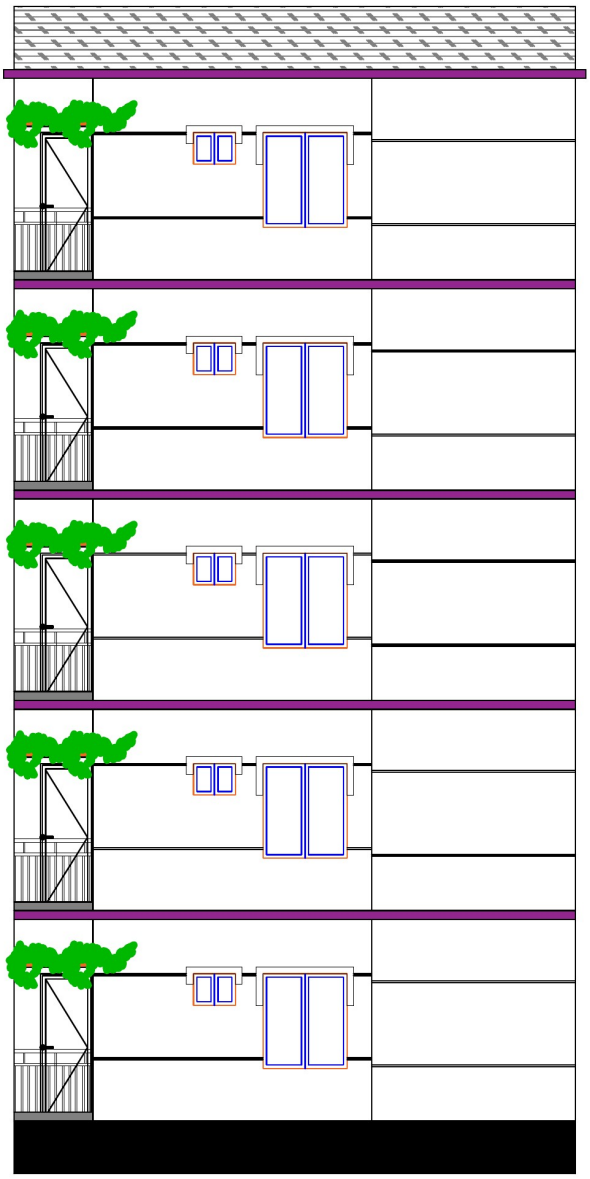
## DESCRIPTION

Elevation of the road-facing facade showing the five storeys, windows and balconies, and the ground floor entrance and parking opening.

## KEY INFORMATION FROM THE DRAWING

- Five storeys (ground + 4 upper floors)
- Window and balcony pattern repeated on each floor

# Back Side Elevation



BACK SIDE ELEVATION

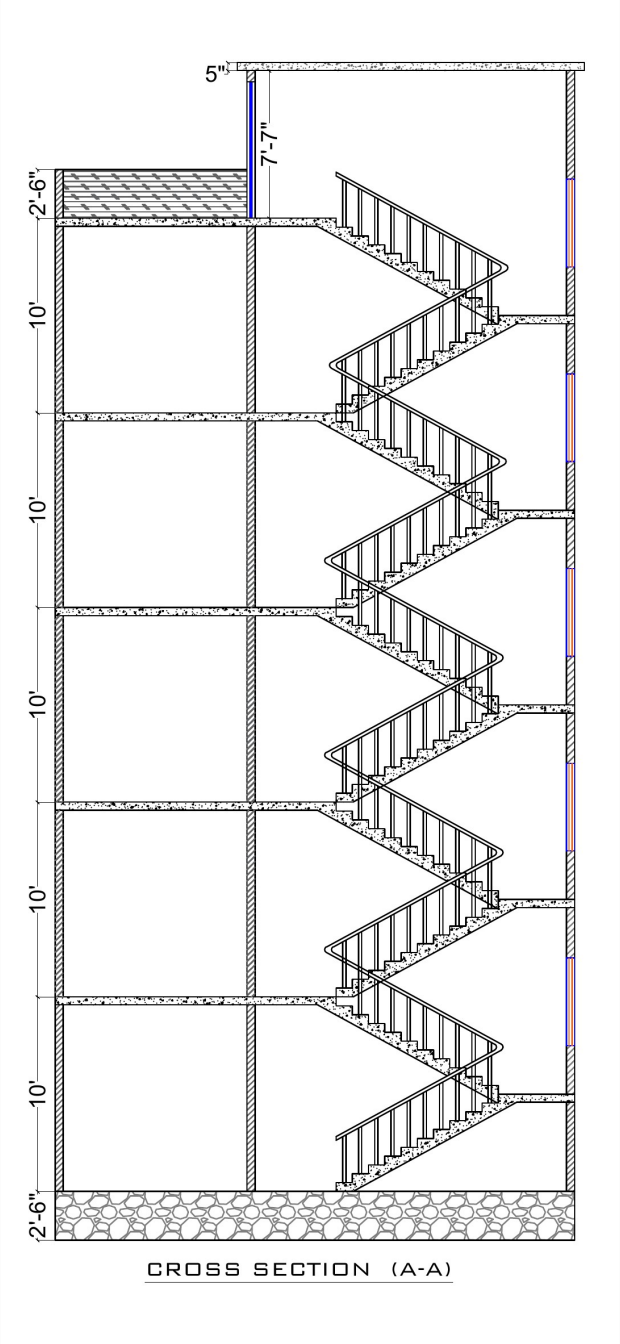
## DESCRIPTION

Elevation of the rear facade showing the five storeys with windows, balconies and planter greenery along the balconies.

## KEY INFORMATION FROM THE DRAWING

- Five storeys, same height as the front elevation
- Balcony and window arrangement of the rear units

# Cross Section (A-A)




CROSS SECTION (A-A)

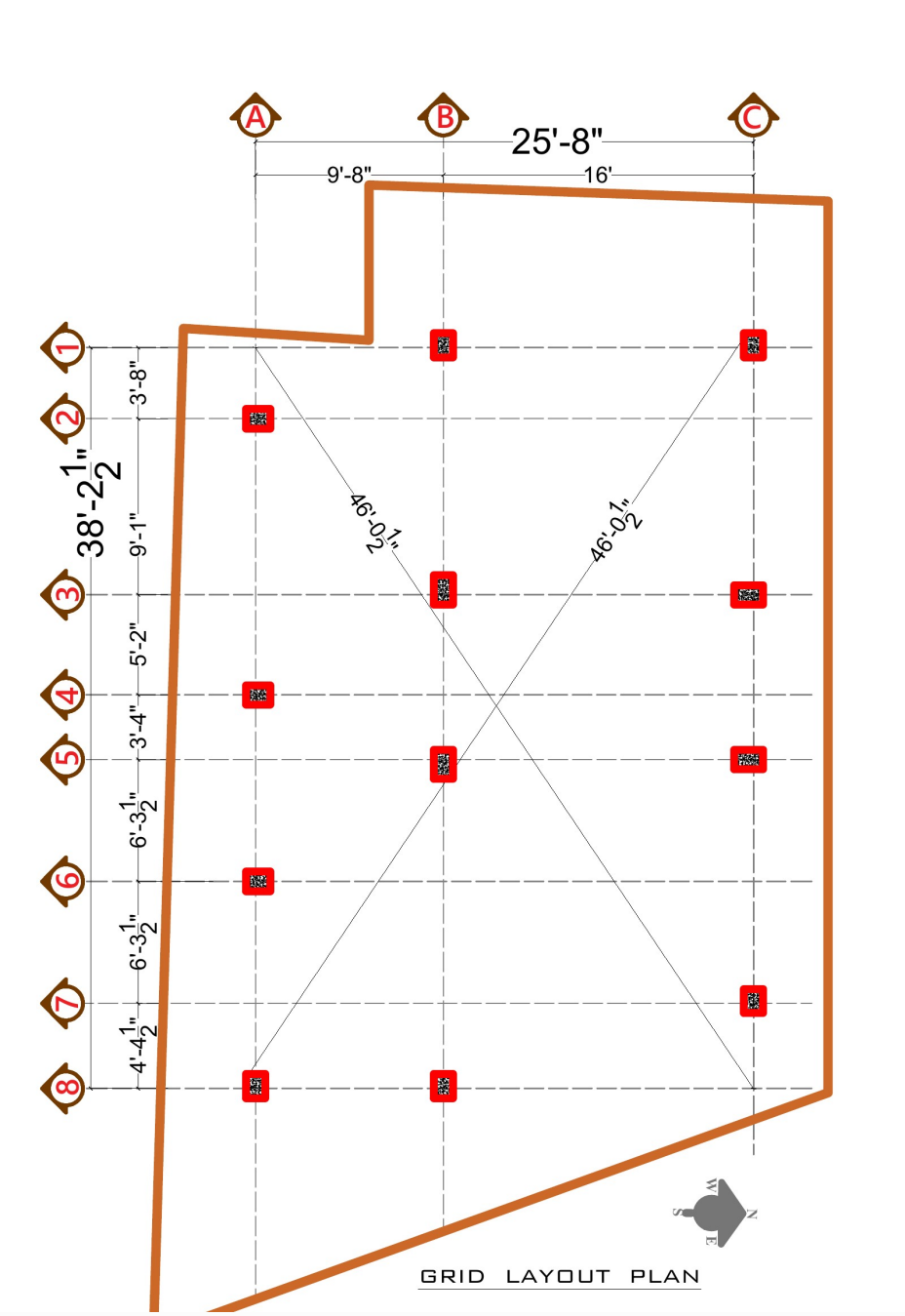
## DESCRIPTION

Vertical section through the building along the stair, showing the floor levels and the stair flights and landings on each floor.

## KEY INFORMATION FROM THE DRAWING

- Floor-to-floor height of 10'-0" as dimensioned
- Stair flights and landings on all five levels
- Roof slab at the top

# Grid Layout Plan



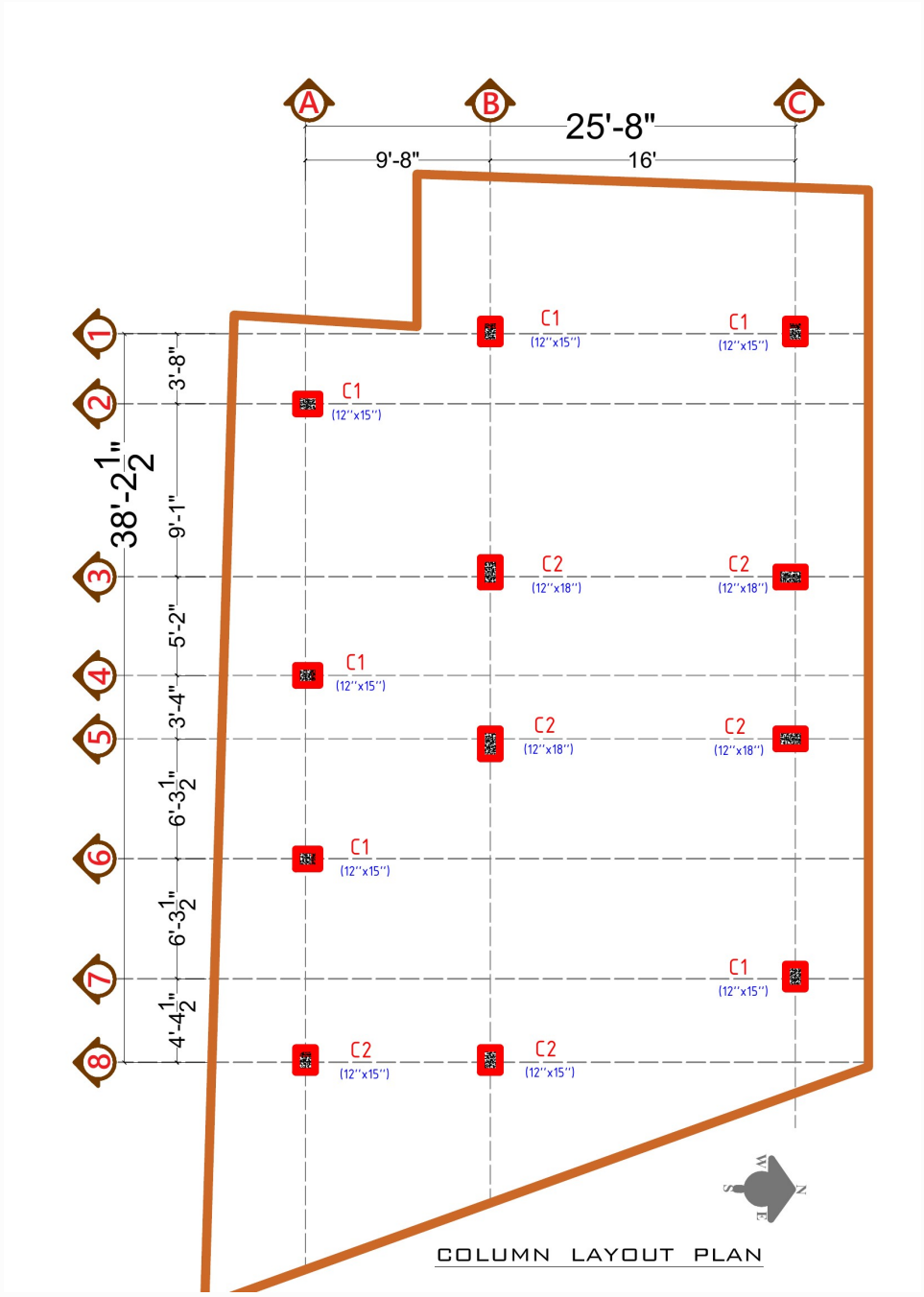

## DESCRIPTION

Column grid used to set out the structure. Gridlines A to C run in one direction and 1 to 8 in the other, with spacing dimensions marked.

## KEY INFORMATION FROM THE DRAWING

- Gridlines A-C and 1-8
- Overall width 25'-8"
- Bay dimensions include 9'-8", 16', 3'-8", 9'-1", 5'-2" and 3'-4"
- Overall depth 38'-2½"

# Column Layout Plan



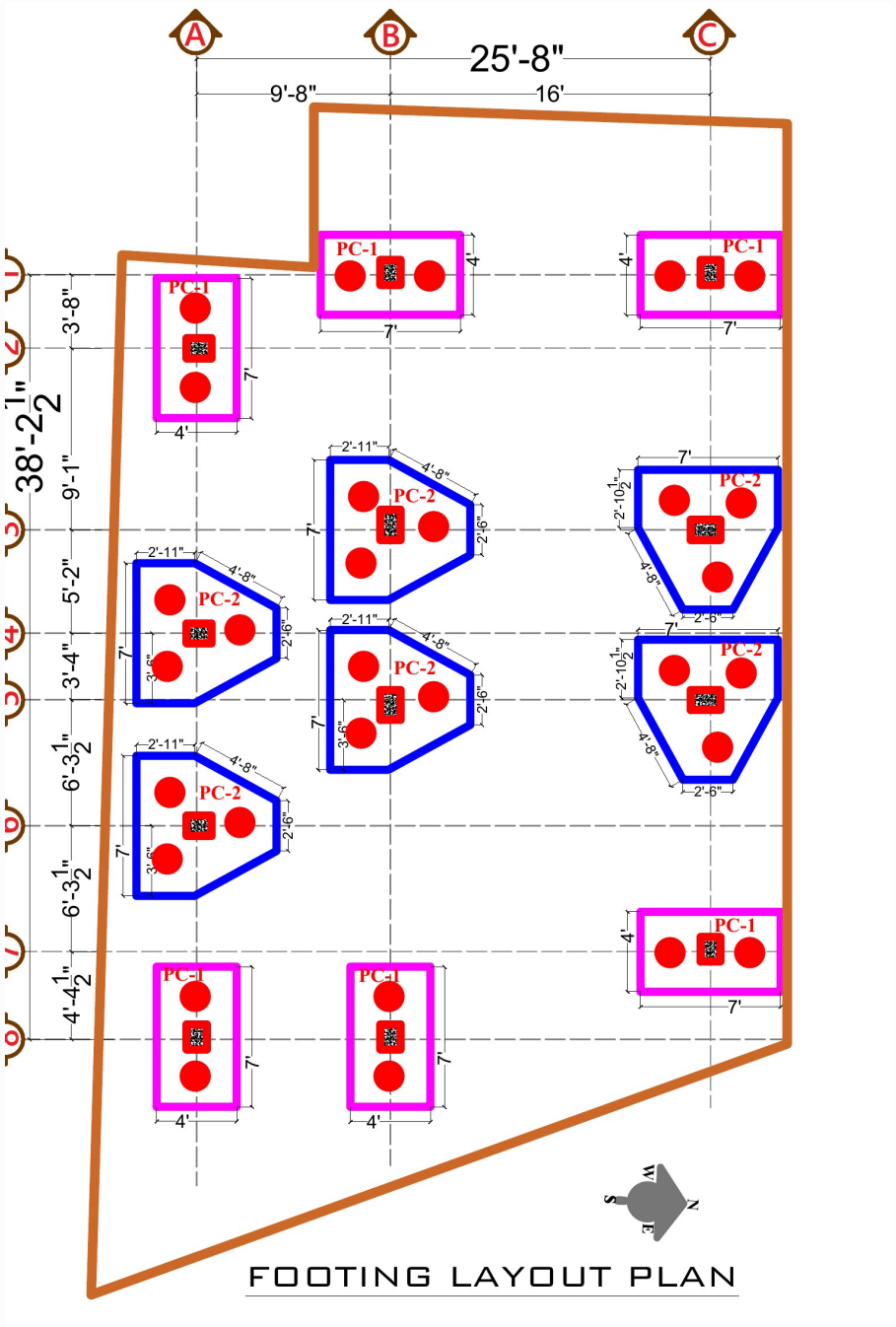

## DESCRIPTION

Locations and marks of all columns on the grid. Two column types are used, with their sizes marked beside each column.

## KEY INFORMATION FROM THE DRAWING

- C1: 12" x 15"
- C2: 12" x 18" (a few marked 12" x 15")
- Reinforcement is on the Column Schedule sheet

# Footing (Pile Cap) Layout Plan



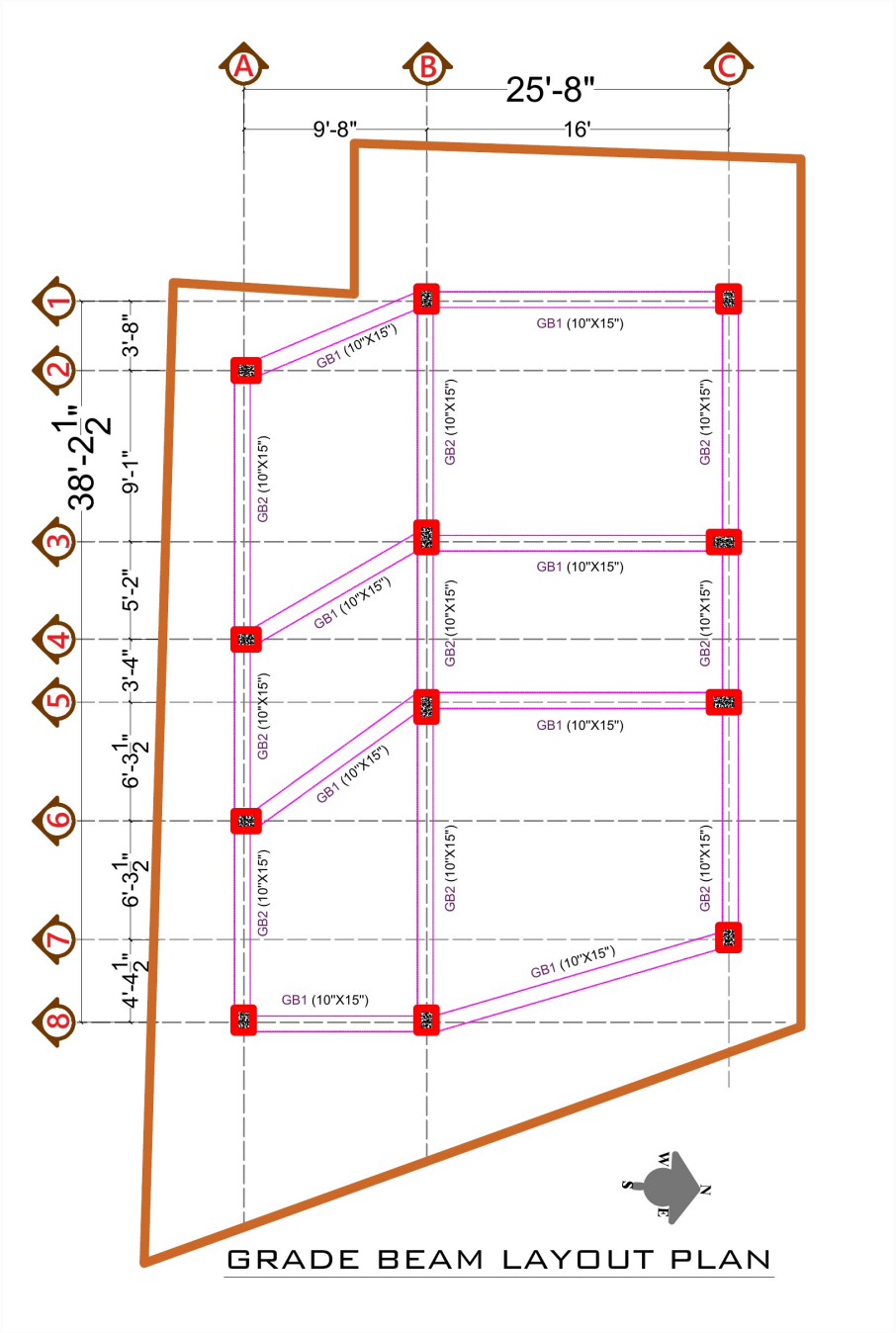

## DESCRIPTION

Foundation layout using pile caps over pile groups. The pile caps are marked by type and the pile positions are shown in each cap.

## KEY INFORMATION FROM THE DRAWING

- Pile cap types PC-1 and PC-2 on the layout
- Pile and pile cap details are on a separate sheet
- Same grid A-C and 1-8

# Grade Beam Layout Plan



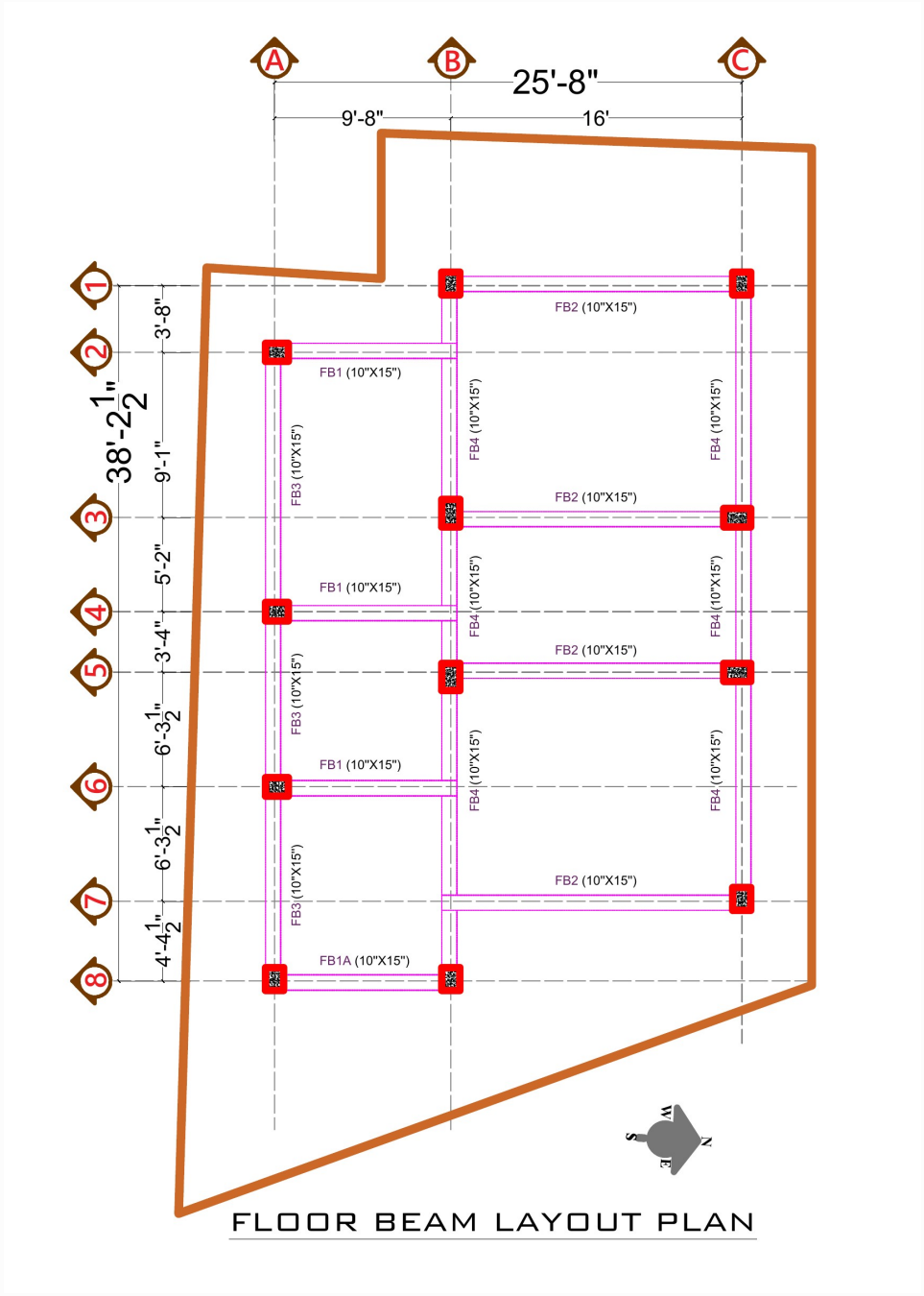

## DESCRIPTION

Plan of the grade beams that connect the pile caps, with a mark and size on each beam.

## KEY INFORMATION FROM THE DRAWING

- Beam marks GB1 and GB2
- All beams 10" x 15"
- Reinforcement is on the grade beam long sections

# Floor Beam Layout Plan



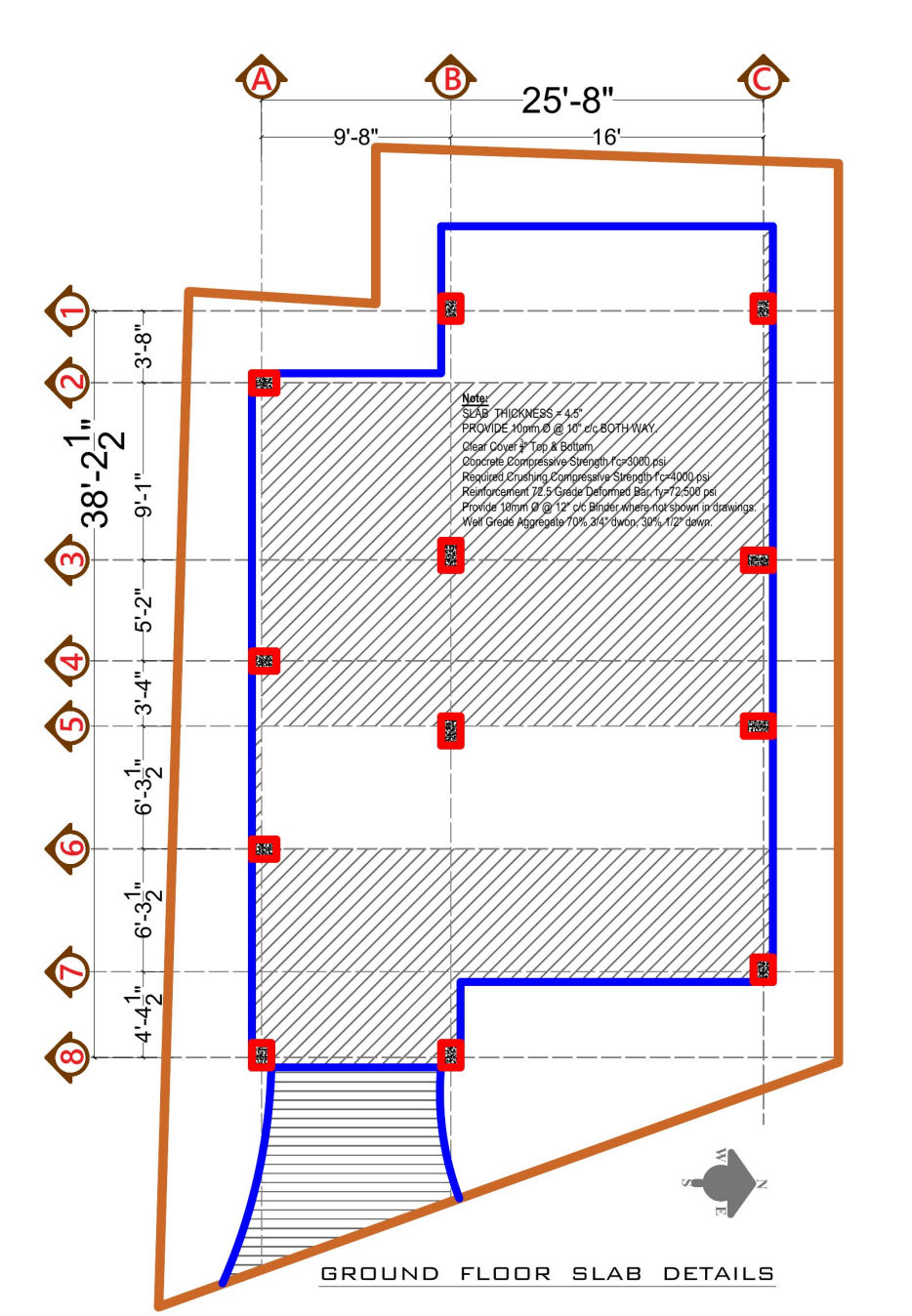

## DESCRIPTION

Layout of the floor-level beams, each marked with its type and size.

## KEY INFORMATION FROM THE DRAWING

- Beam marks FB1, FB1A, FB2, FB3 and FB4
- All beams 10" x 15"
- Same grid and 25'-8" overall width

# Ground Floor Slab Details



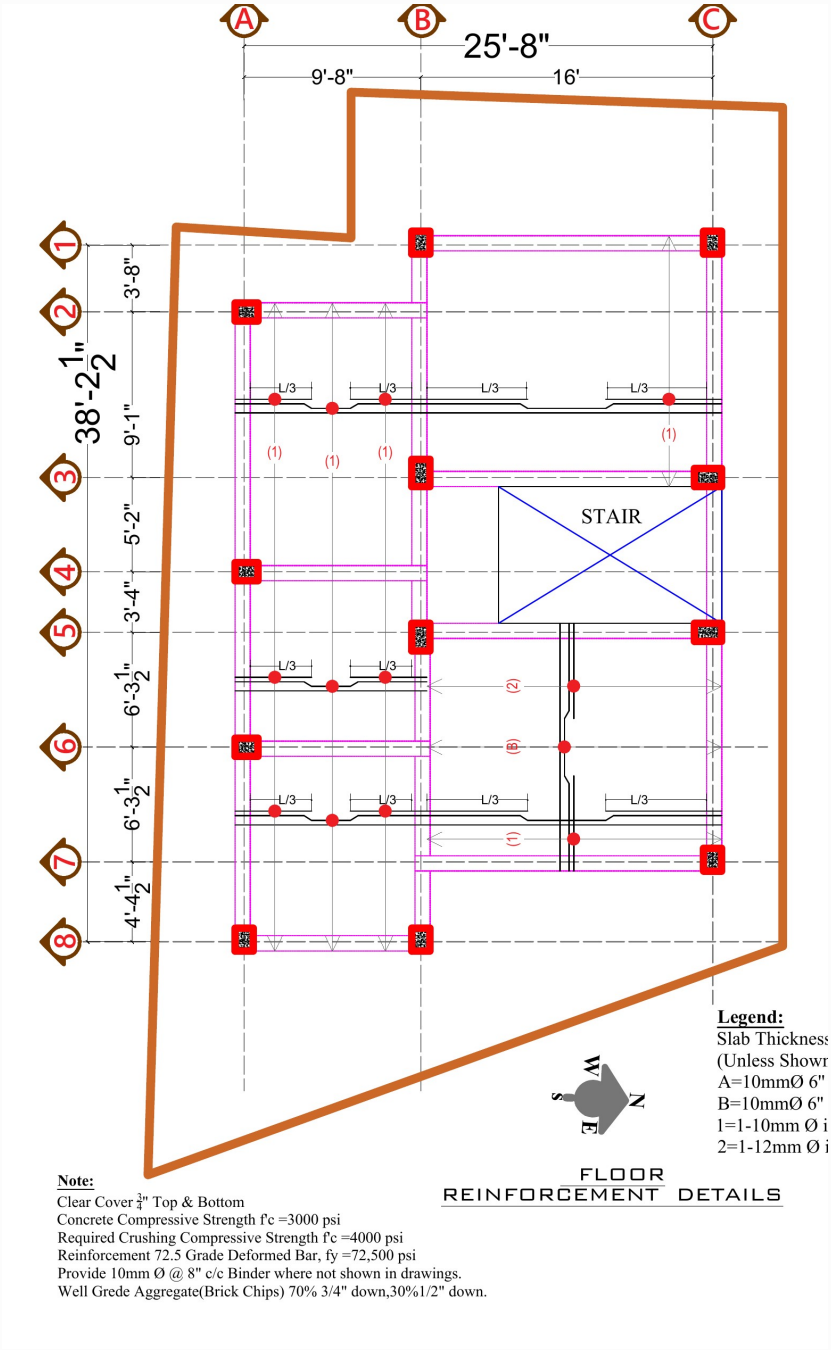

## DESCRIPTION

Plan of the ground floor slab with the general notes for slab thickness, reinforcement and materials.

## KEY INFORMATION FROM THE DRAWING

- Slab thickness 4.5"
- 10 mm dia bars @ 10" c/c both ways
- Clear cover 3/4" top and bottom
- Concrete f'c 3000 psi; required average strength 4000 psi
- Reinforcement 72.5 grade, fy 72,500 psi; binder 10 mm dia @ 12" c/c

# Floor Reinforcement Details



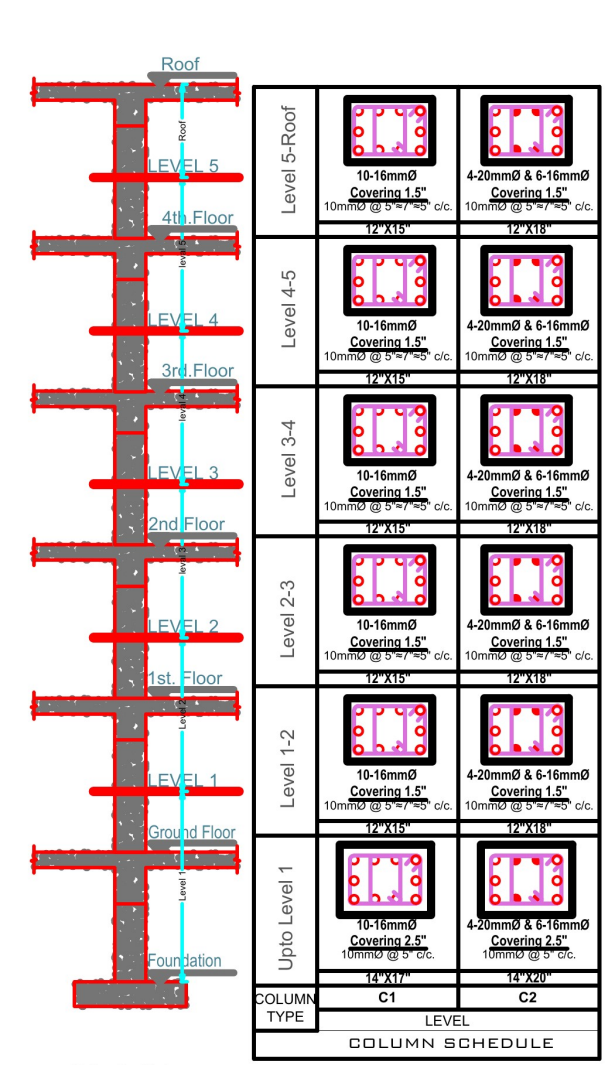

## DESCRIPTION

Reinforcement layout for the typical floor slab, with the stair opening and L/3 zone marks, and a legend giving the bar types.

## KEY INFORMATION FROM THE DRAWING

- Slab thickness 5" unless shown otherwise
- Legend: A and B = 10 mm dia @ 6" c/c alternate cranked; 1 = 1-10 mm dia between cranked bars; 2 = 1-12 mm dia between cranked bars
- Clear cover 3/4" top and bottom
- Binder 10 mm dia @ 8" c/c where not shown

# Column Schedule




**Notes for Column:**

1. Concrete design strength, f'c = 3500 psi.
2. Required average strength, f'c = 4500 psi.
3. All Reinforcement bars should be deformed bar having yield strength, fy = 72,500 psi.(72.5 Grade)
4. Concrete mix ratio (1:1.5:3) confirm by trial mix ratio
5. If any information is not given should be taken from General Notes.
6. Well Graded Course Aggregate(70% 3/4" Down, 30% 1/2" Down).

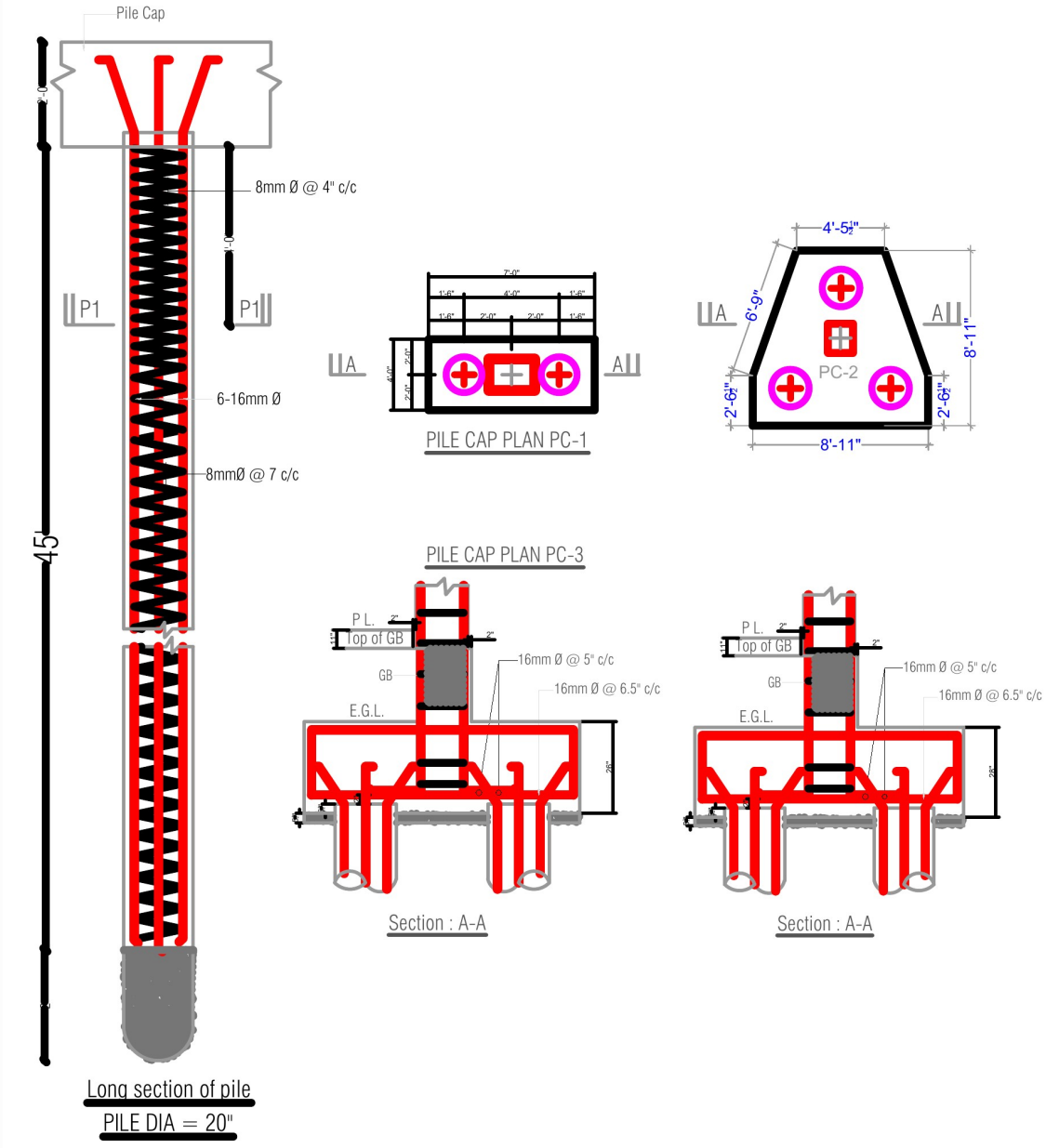
## DESCRIPTION

Table of the column types by level, from the foundation to the roof. For each level it gives the section size, the longitudinal bars, the ties and the cover.

## KEY INFORMATION FROM THE DRAWING

- C1 12" x 15" and C2 12" x 18", increasing to 14" x 17" and 14" x 20" at the lower level
- Bars 10-16 mm dia for C1; 4-20 mm dia with 6-16 mm dia for C2
- Ties 10 mm dia @ 5" to 7" c/c
- Concrete f'c 3500 psi, required average 4500 psi, mix 1:1.5:3
- Reinforcement 72.5 grade, fy 72,500 psi

# Pile and Pile Cap Details



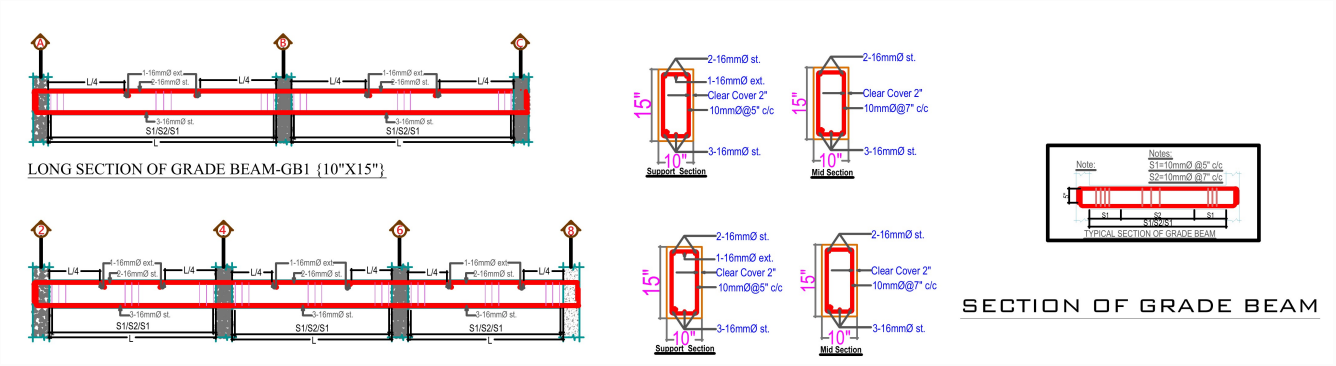

## DESCRIPTION

Long section of the pile, plans of pile caps PC-1, PC-2 and PC-3, and the sections through the caps with their bars and levels.

## KEY INFORMATION FROM THE DRAWING

- Pile diameter 20", with a 45' length dimension
- Pile ties 8 mm dia, 4" c/c
- Pile cap bars 16 mm dia @ 6.5" c/c and @ 5" c/c, with 6-16 mm dia bars
- Pile cap depths of 28" and 26" shown on the sections

# Long Sections of Grade Beams GB1 and GB2



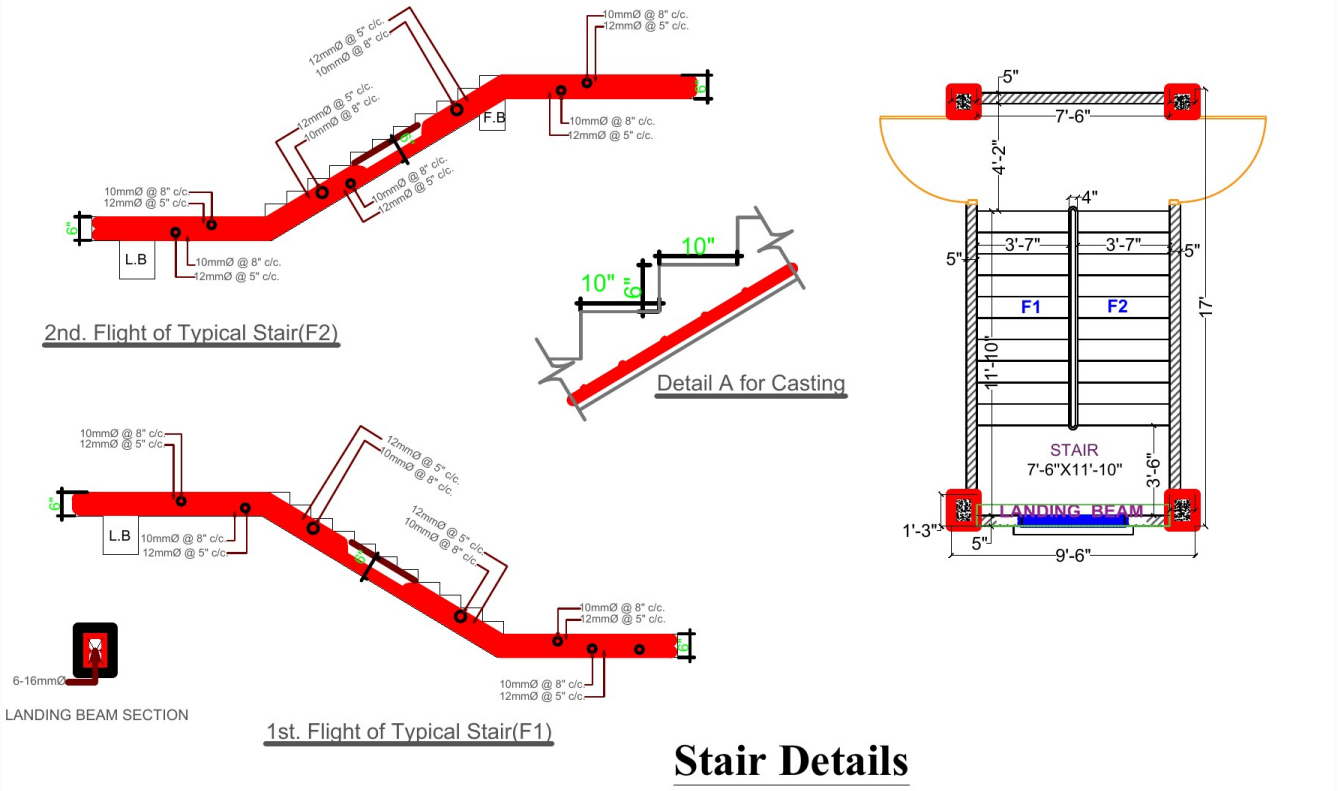

## DESCRIPTION

Longitudinal sections of grade beams GB1 and GB2 with the support and mid section of each, showing the main bars, extra bars and stirrup spacing.

## KEY INFORMATION FROM THE DRAWING

- Beam size 10" x 15", clear cover 2"
- Straight bars 2-16 mm dia and 3-16 mm dia with 1-16 mm dia extra bars near supports (L/4)
- Stirrups 10 mm dia @ 5" c/c at supports and @ 7" c/c in the middle

# Stair Details



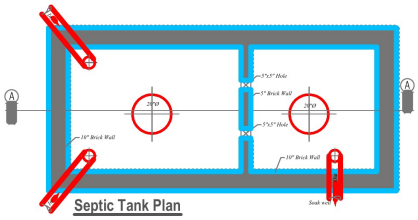

## DESCRIPTION

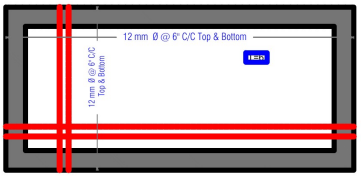
Typical stair details: the first and second flights with their reinforcement, the casting detail, the landing beam section and the stair plan.

## KEY INFORMATION FROM THE DRAWING

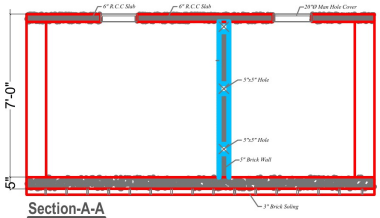
- Flights F1 and F2; stair 7'-6" x 11'-10"
- Main bars 12 mm dia @ 5" c/c, distribution 10 mm dia @ 8" c/c
- Landing beam with 6-16 mm dia bars

# Septic Tank, Soak Well and Underground Reservoir Details

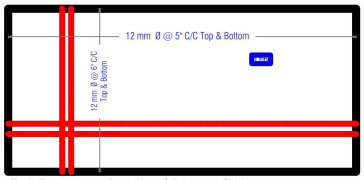


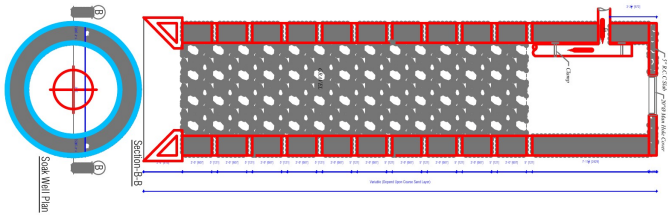

REINF. DETAIL'S OF SEPTIC TANK TOP SLAB



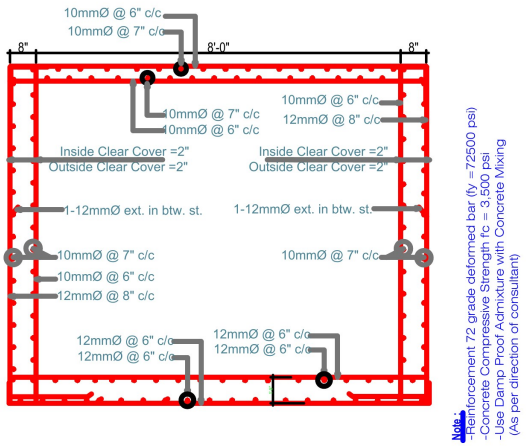

Section-A-A




Reinforcement Details of Bottom Slab.




**DETAIL OF SEPTIC TANK**




**Section Of Under Ground Water Reservoir**

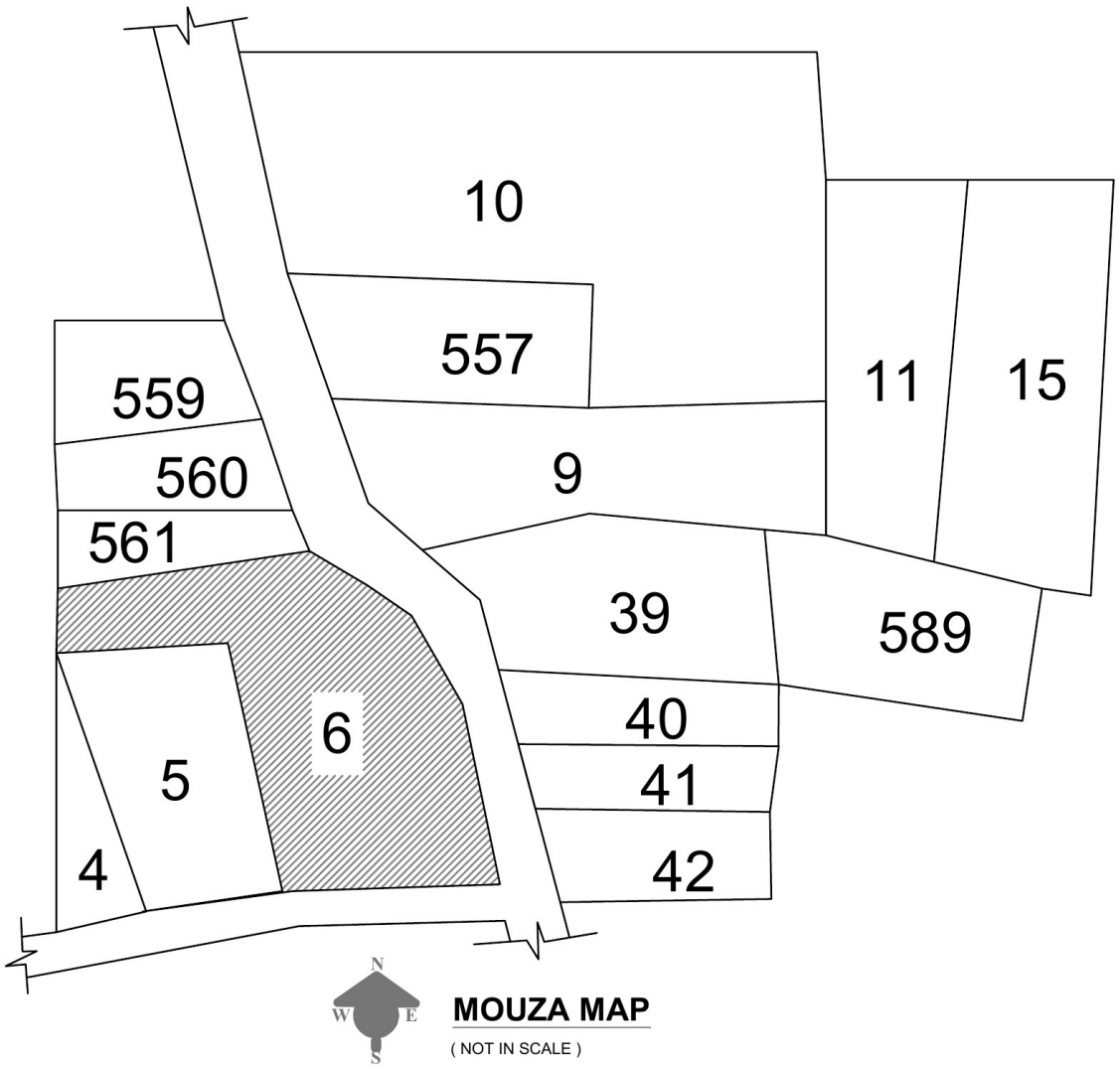
## DESCRIPTION

Plans, sections and reinforcement details for the septic tank and soak well, and the section of the underground water reservoir.

## KEY INFORMATION FROM THE DRAWING

- Septic tank top and bottom slabs 6" thick with 12 mm dia bars
- Brick walls 5" and 10" thick
- Reinforcement 72 grade, fy 72,500 psi; concrete f'c 3,500 psi
- Damp-proof admixture in concrete, clear cover 2"

# Mouza Map



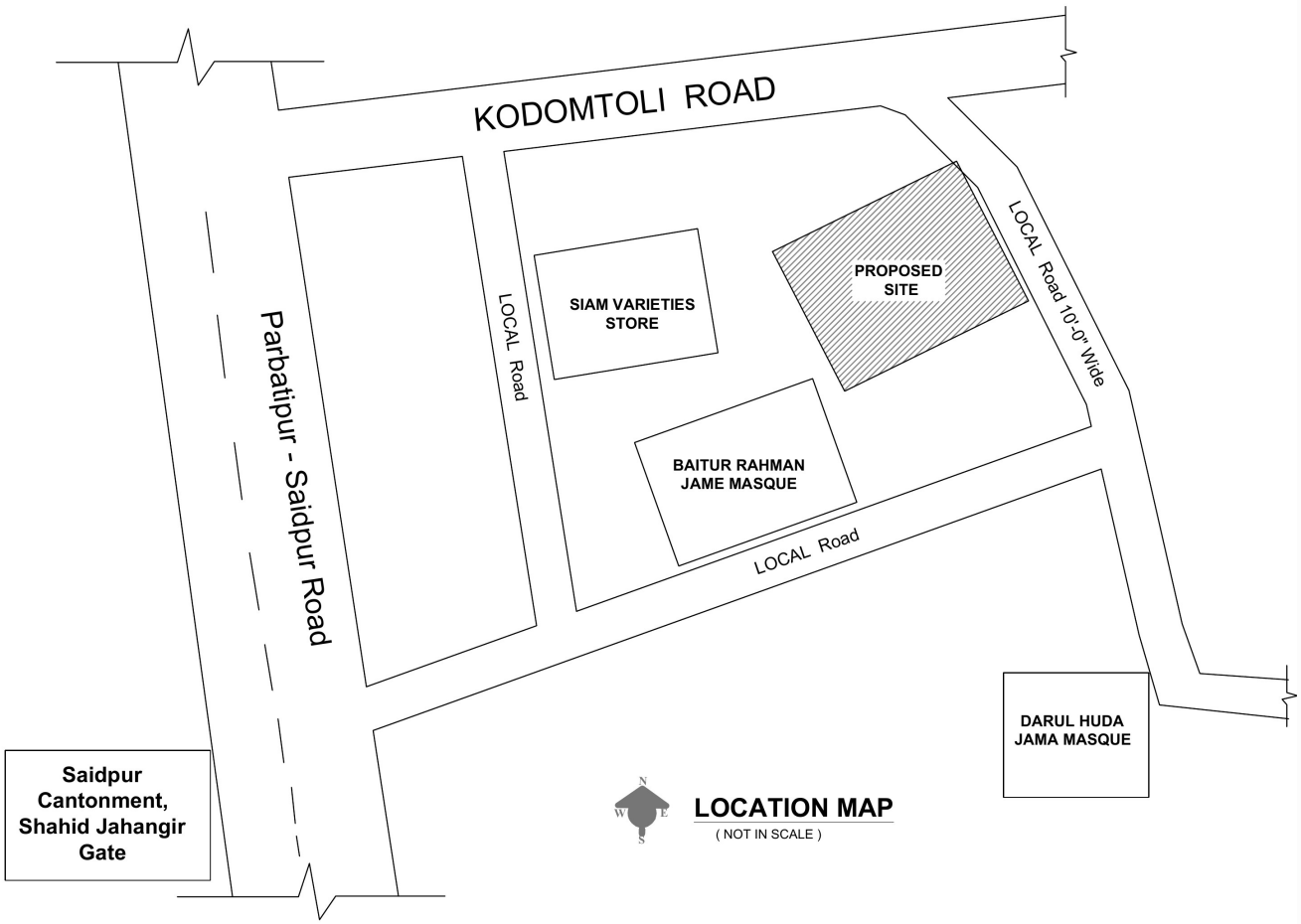

## DESCRIPTION

Cadastral (mouza) map of the surrounding plots with dag numbers. The project plot, Dag 6, is shaded. Not to scale.

## KEY INFORMATION FROM THE DRAWING

- Project plot Dag No. 6
- Adjoining dags 4, 5, 9, 10, 11, 15, 39, 40, 41, 42, 557, 559, 560, 561 and 589
- Not drawn to scale

# Location Map




## DESCRIPTION

Sketch map showing the proposed site in relation to nearby roads and landmarks. Not to scale.

## KEY INFORMATION FROM THE DRAWING

- Kodomtoli Road and Parbatipur-Saidpur Road nearby, with local roads
- Baitur Rahman Jame Masque, Siam Varieties Store and Darul Huda Jama Masque nearby
- Saidpur Cantonment, Shahid Jahangir Gate to the west
- Not drawn to scale